How much leaf is left?

A loop goes somewhere. · Curiosity day

Name _____ Date _____



A drawing model for a journey. Use the invented table distances, not the picture's size or dot spacing.

PRETEND JOURNEY DISTANCES · MILLIMETERS

Part	Whole route	First stretch	Next stretch
Length (mm)	32	12	9

Use the table numbers. The picture is not a ruler.

1. Walk 12 mm, then 9 mm. How far in all?

$12 + 9 =$ _____ Show your thinking.

2. Route is 32 mm. After 12 mm and 9 mm, how much is left? Show both steps.

$32 - (12 + 9) =$ _____ Show your thinking.

3. Add 8 mm to the 32 mm route. After a 21 mm walk, how much is left?

$(32 + 8) - 21 =$ _____ Show your thinking.

4. A trip goes 7 mm, then 6 mm, then 5 mm. How far in all?

$7 + 6 + 5 =$ _____ Show your thinking.

5. Trip A: 12+9 mm. Trip B: 7+6+5 mm. Which is longer, and by how much?

$(12 + 9) - (7 + 6 + 5) =$ _____ Show your thinking.

6. 11 mm left. Two friends share it as evenly as they can. How much each, and how much left over?

$11 \div 2 =$ _____ Show your thinking.

The arch begins

Little spots a green arch on the leaf. "A tiny bridge!" The arch flattens, and a small head reaches past a vein. Big watches from the path.

The rear catches up

Front grips hold the leaf. The rear slides closer and the long middle lifts into a tall loop.

Hold, then reach

The prolegs anchor the rear: they hold it in place. The front stretches ahead as the body straightens.

A clue on the leaf

Little draws the head before a vein fork, then after it on the same leaf vein. The two marks are evidence — clues that can support an idea.

A pretend leaf route

"Does every loop go one inch?" Little asks. The name sounds like a ruler, but real steps vary.

A later shape

Little imagines later chapters. The caterpillar becomes a pupa, then a moth. That change of body form is metamorphosis, and it takes time.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

A loop goes somewhere.

Explain a tiny grip.

A loop goes somewhere. · Curiosity day

Name _____ Date _____

Choose a scene. Write two linked sentences: one observation, then an explanation using because. Draw a detail that supports your idea.



The rear catches up



A grip, then a reach



The leaf keeps a clue

Draw the detail you noticed. Add an arrow to show movement or a dot to show position.

prolegs · anchor · loop · reach · evidence · inference · because

The middle rises because … What happens when the rear gets closer to the gripping front?

The rear grip matters because … What keeps the body in place while the front reaches?

I infer that it moved because … Compare the head with the same vein fork in both sketches.

Reread your words. Does the detail support your explanation? Check the first capital and each ending mark.

A loop goes somewhere.

01 Why does the middle rise into a loop?



Front grips; rear comes closer. Notice the long legless middle rather than counting the partly hidden feet.

Look closer. Find the head at the right. Follow the body back to the rear feet. Where is there empty space underneath?

A little explanation. The front legs hold the leaf while the rear comes closer. The long middle bends upward into a loop. Inchworms have jointed legs near the head and fleshy gripping parts called prolegs near the rear, with a long gap between. They do not walk on a row of legs along that middle.

Try this. Draw a long curved line with its two ends close together. Imagine the front end stays in one place as the rear comes closer. Which part must bend?

02 What holds on while the head reaches?



Rear grips; front reaches. This simplified still picture shows one moment in a repeating movement.

Look closer. The front is lifted. Follow the body to the far left. Which feet still touch the leaf?

A little explanation. Now the rear prolegs anchor the body: they hold it in place. The front reaches ahead as the middle straightens. When the front legs grip again, the rear can catch up. The two ends take turns holding on. That repeating sequence carries the caterpillar across a leaf.

Try this. Look at the rear grip. What might happen if it slipped while the front was still reaching? Use a detail from the picture to explain.

03 How could a leaf vein show it moved?



Two imagined observations of the same leaf. Compare the head marks against the same fork.

Look closer. Compare the two leaf sketches. Each dot records the head. Find the same Y-shaped vein fork in both. What changed?

A little explanation. A bent body alone does not tell us how far the animal traveled. The first head mark is before the fork; the later mark is beyond it. The fork is a reference point, a place to compare positions against. Those changed positions are evidence for movement. They do not tell us its speed without times.

Try this. Draw a Y and put a dot before it. Draw the same Y again with a dot beyond it. Explain how the two drawings support the idea that the head moved.

04 Will this caterpillar always crawl?



An imagined later life stage. The notebook skips time; this change does not happen during one evening.

Look closer. Find three life stages in the notebook. How does the last shape differ from the first?

A little explanation. An inchworm is a caterpillar, the young stage of a geometer moth. It grows, becomes a pupa, then changes into a moth. This big change in body form is metamorphosis. It takes time; this scene imagines a later stage. A moth has six jointed legs. The caterpillar's fleshy prolegs do not remain.

Try this. Follow the notebook arrows: caterpillar, pupa, moth. Name one body feature that changes. What extra observation would help you learn how long the change takes?